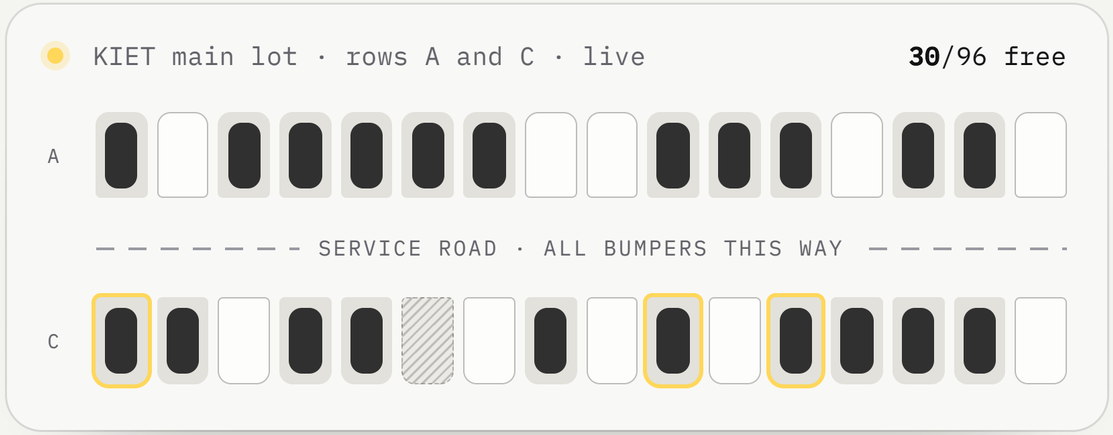


INNOTECH'26 · 30 October 2026

Category 6 · Gen Z Budding Project Solutions

Dept. of IT & CSE (Cyber Security), KIET Deemed to be University

Team Arrivo

Krish Singh · Kritika Sakhuja · Prisha Soam · Vaibhav Kumar

arrivo.

Park like you booked a seat.

press → to begin

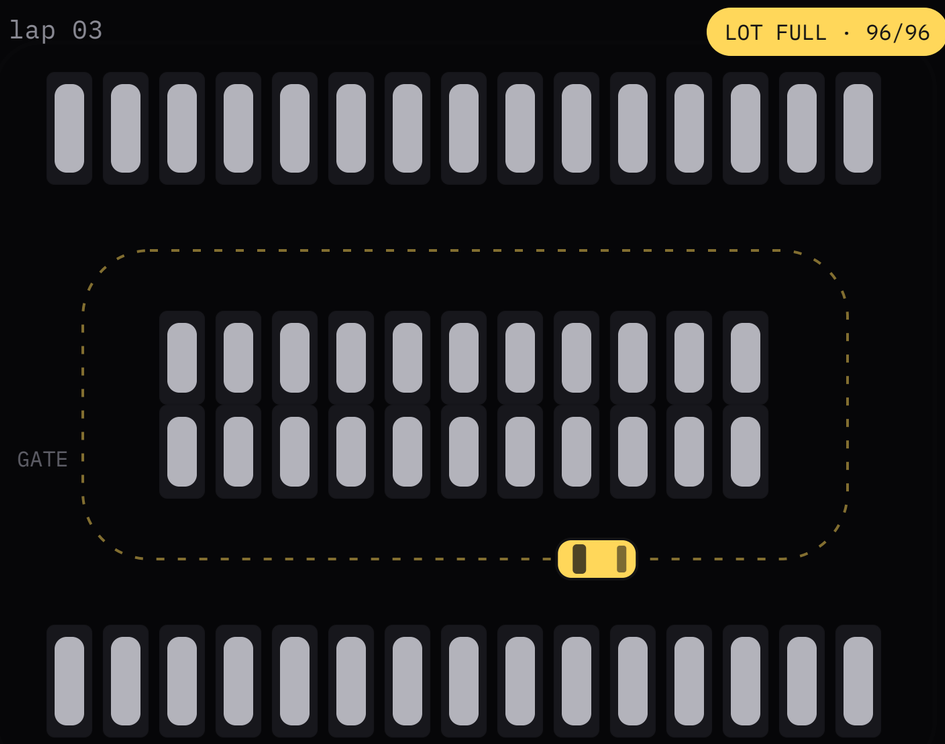
Title

01 —— Every weekday morning

09:58

The lot is full. You just don't know it yet.

So you circle. Row A, row C, back past the gate. The lecture starts without you, and the car behind you is doing the same lap.



30%

of traffic in busy city centres is drivers cruising for a parking space.

Shoup, "Cruising for parking", 2006

17 h

a year the average driver spends hunting for parking.

INRIX parking study, 2017

0

ways to know which of the 96 bays at KIET is free before you drive in. Until now.

our count of the KIET main lot

02 —— Why it keeps happening

First come, first guessed.

The problem isn't too few bays. It's that nobody can see them, book them, or plan them.



01

No visibility

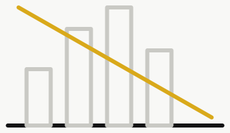
From the gate you can't tell which of 96 bays is empty, so you go and look.



02

No booking

Seats, trains and tables can all be booked. A 2.5 × 5 m bay still can't.



03

Blind operators

Nobody knows which rows fill first, or when, so nothing gets planned.



04

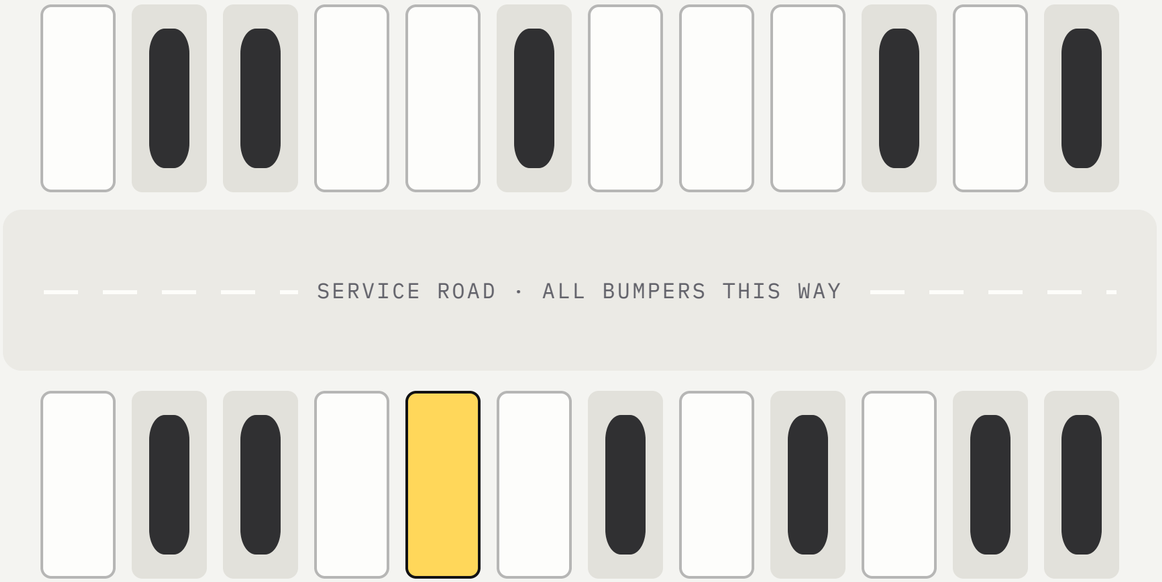
Wasted minutes

Every lap burns fuel and time, for you and for the queue behind you.

03 —— The idea

Book a bay the way you book a seat.

Ticket apps made picking your exact seat feel normal: you see the hall, you pick the empty one, it's held for you. Arrivo brings that to parking. You see every bay on a live plan, pick a free one, choose your hours, and it's yours when you arrive.



a parking lot · 24 bays · **1 is yours**

same idea, new place.

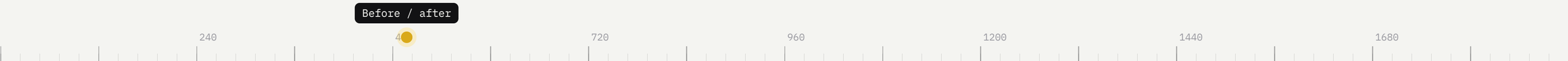
04 —— What changes

Before and after.

	WITHOUT ARRIVO	WITH ARRIVO
Finding a bay	Drive in, circle, look	Tap a free bay on a live plan
Knowing it'll be free	Hope	Held for your hours
Running late	Someone else took it	15-minute grace window
Two cars, one bay	An argument at the gate	Impossible: the database refuses it
The lot's owner	Guesses how it's used	Every booking is data

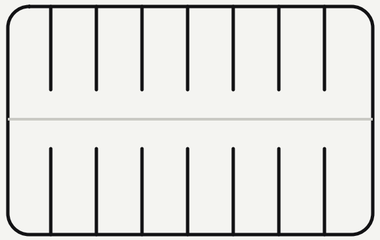
What's new here

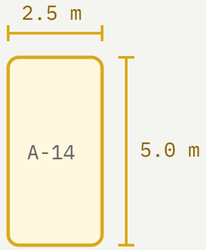
- **Bay-level booking**
- **X-ray lens** over a live plan
- A lot **traced from its satellite photo**
- **Live** for every viewer at once

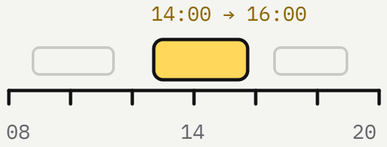


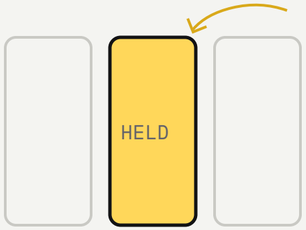
05 — How a booking works

Five steps, lot to parked.











01

A lot.

Each lot is drawn once, row by row, from its satellite photo.

02

A bay.

Every bay has a name, a size and a type: standard, compact, EV or step-free.

03

A window.

Pick when you arrive and how long you stay. Taken hours show grey.

04

A hold.

Confirm, and the bay is held in your name with a bay pass and a code.

05

You arrive.

Drive straight to your bay. Nobody else could book it.

06 —— Built on the real lot

Drawn from the real KIET lot.

We traced the lot beside the KIET football ground from its satellite photo. Every bay sits on the photo's own pixels.

96

bays, all bookable

5

rows, A to E

2.5 × 4.9 m

a standard bay

7.7 px/m

50 m on the photo's scale bar = 385 px



Source: Google Earth capture of KIET, Ghaziabad. Imagery © Google.

The real lot

240 480 720 960 1200 1440 1680

07 — The live plan

Tap it, tilt it, or pick up the lens.



3D view · buildings rise out of the plan



Dark mode · same plan, at night



X-ray lens · hold & drag over the plan

Live plan

240 480 720 960 1200 1440 1680

08 — Booking

Pick your hours. Get a pass.

Choose when you arrive and how long you stay, add your number plate, and hold the bay. About a minute, start to finish.

01 Clashes are crossed out

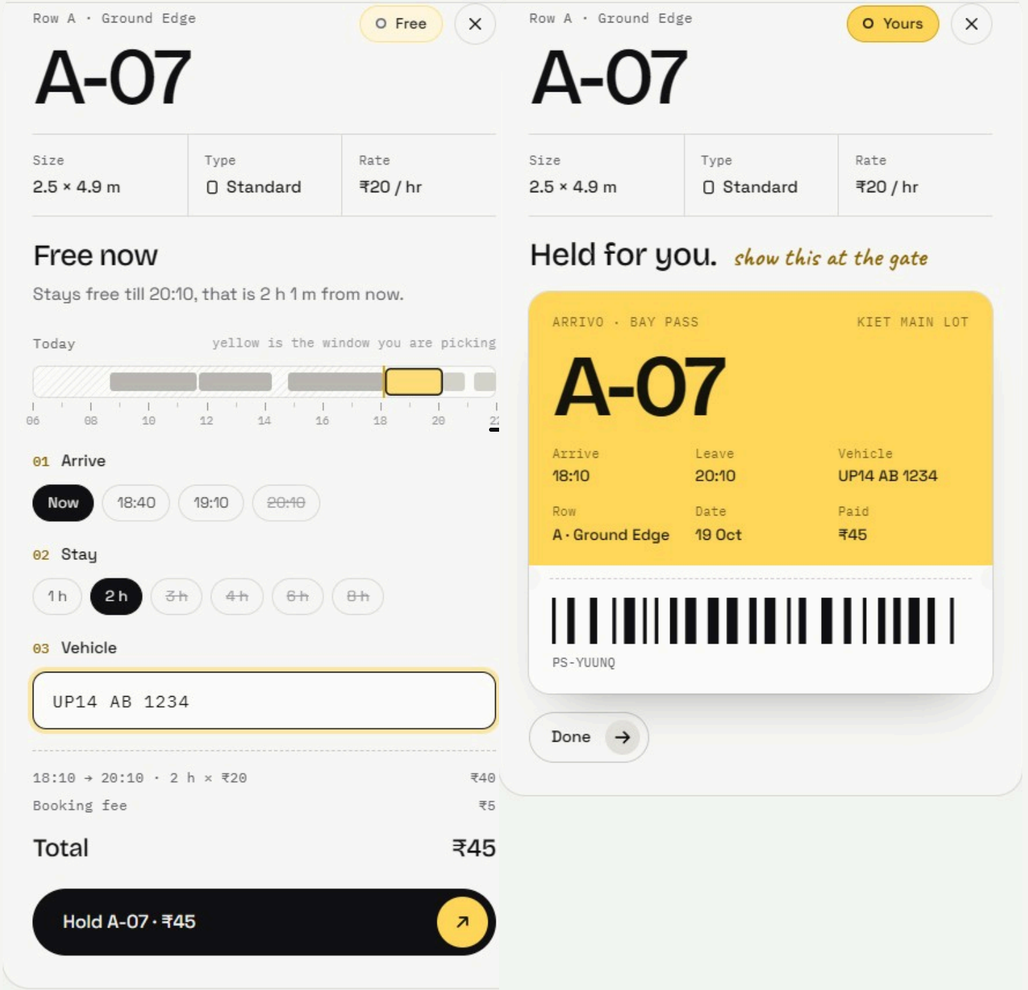
Times that overlap someone else's booking can't be picked.

02 A pass with a code

Show it at the gate. It survives a refresh and opens in any tab.

03 Late? 15 minutes' grace

After that the bay opens up for someone else.



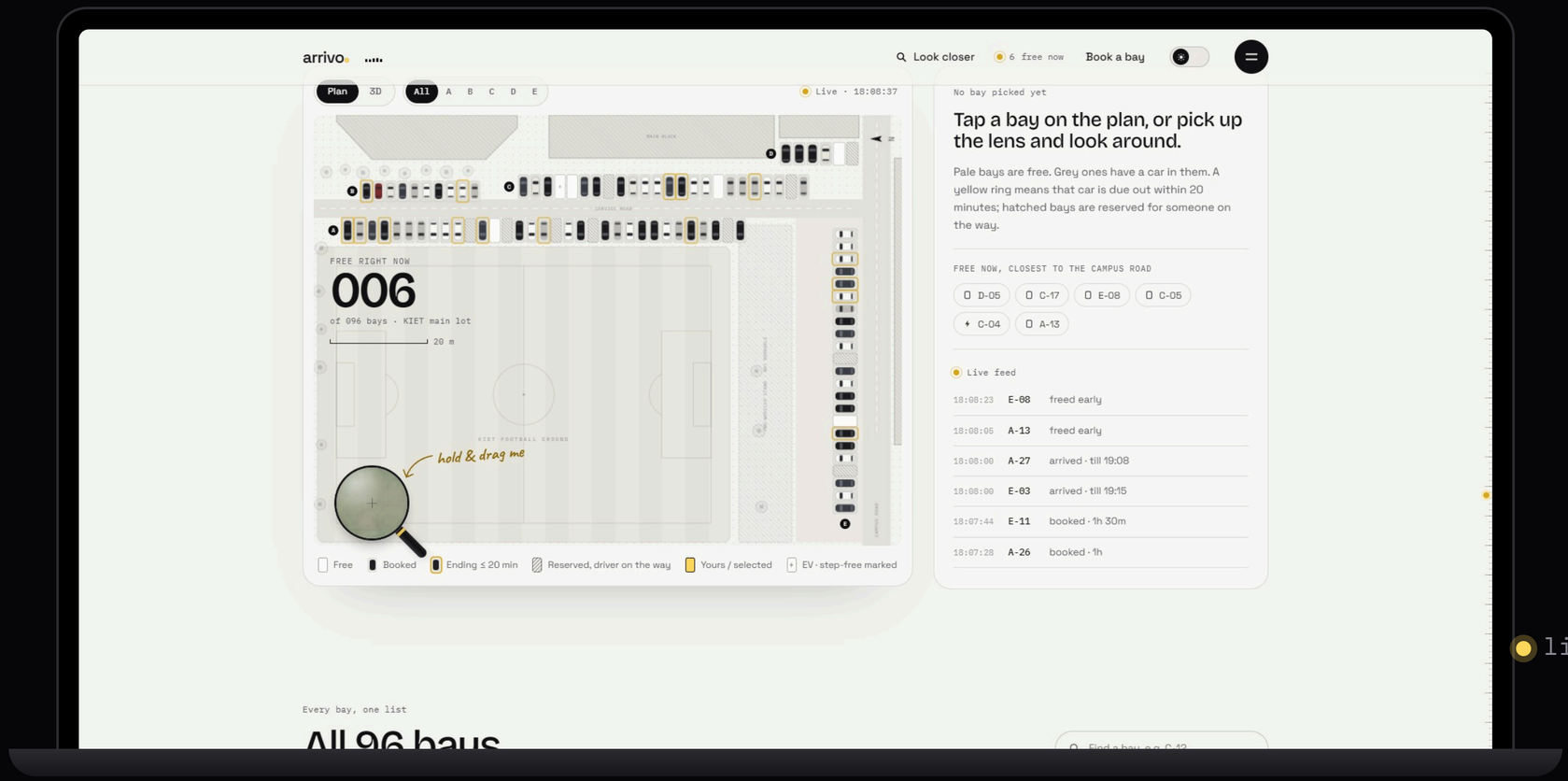
nobody else can take it.

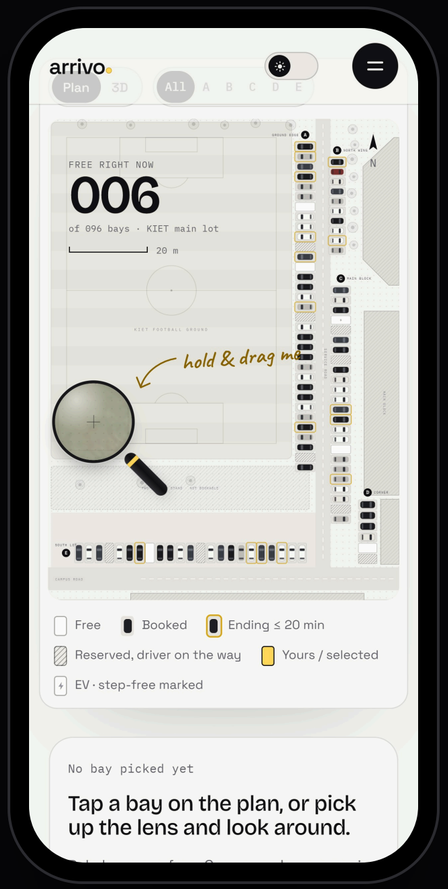
Booking

240 480 720 960 1200 1440 1680

09 —— Demonstration

Now, live.





live demo on the KIET lot, during the talk

Open the live site L

Live demo

240 480 720 960 1200 1440 1680

10 — Tech stack

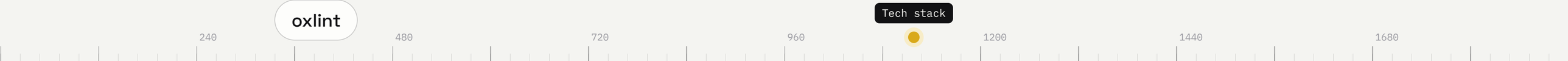
What it's built with.

Languages	JavaScript · Python · HTML · CSS · SQL
Frontend	React 19 live map & panel · Vite 8 build · Motion animation · Lenis smooth scroll · SVG the plan, by hand
Backend	Python 3 standard library only · http.server · sqlite3 · threading · Server-Sent Events live push
Design	Bricolage Grotesque · Space Grotesk · IBM Plex Mono · Caveat · paper · ink · one yellow
Tools	Google Earth lot capture · Cloudflare Tunnel share link · Vite dev server · oxlint

7,400
lines of frontend code

1,121
lines of Python, in one file

0
pip installs for the backend



11 — Architecture

How a booking travels.

Browser

Your phone

Live plan, lens, panel. Taps a free bay and books it.

POST /api/bookings

server.py

Python server

Checks the rules, saves the booking, tells every open page. Also runs demo cars so the lot looks alive.

INSERT · one locked s

parkseat.db

SQLite

Bookings and the live feed. A trigger refuses any two bookings that overlap on one bay.

live push (SSE)

Browser

A friend's laptop

Sees that bay turn booked the moment it happens.

same push, every viewer

1 tap a bay → 2 POST the booking → 3 save, or refuse a clash (409) → 4 push to every screen

12 —— Under the hood

Four rules that keep it honest.

01 · no double booking

Two cars can never get one bay.

Checked inside a locked transaction, and again by a database trigger.

```
# two windows overlap when
if a.start < b.end and b.start < a.end:
    raise ApiError(409, "overlap")
```

02 · status from time

Every bay knows what it is.

Worked out from the clock and the day's bookings, never typed in by hand.

free	nothing booked now
booked	a car is in it
ending	due out in ≤ 20 min
reserved	due in within 30 min
yours	booked from this device

03 · live without refresh

Every screen updates at once.

One open connection per page. If it drops, the page reconnects and catches up.

```
Content-Type: text/event-stream

event: change
data: {"booking": {"bayId": "A-07", …}}
```

04 · private by default

No sign-up. No plates on show.

A random device ID marks your bookings, and only its hash is stored.

```
owner = sha256(device_id)

# other people's plates
"UP14 AB 4821"  →  "UP14 •• 4821"
```



13 —— Impact

Less circling, cleaner air.

11 SDG

Sustainable cities and communities

Less traffic crawling round the lot and the gate, and a fair, first-booked system instead of first-lucky.

13 SDG

Climate action

Every lap you don't drive is fuel you don't burn. A booked bay means one straight drive in.

9 SDG

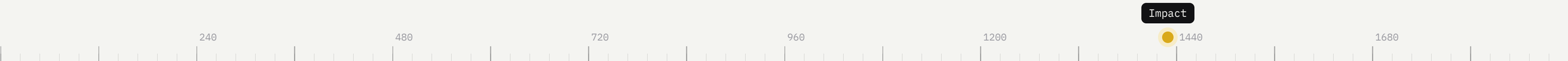
Industry, innovation and infrastructure

Gets more out of the lot that already exists: no new concrete, just a better way to use it.

---

500 engine-minutes saved a day, on one lot, if 100 drivers each skip 5 minutes of circling.

illustrative estimate



14 —— Where it goes next

One lot today. Every lot after.

Same method, any lot

Trace it once, book it every day.

Adding a lot means drawing its rows on a satellite photo, like we did for KIET.

- Campuses
- Hospitals
- Malls
- Metro stations
- Offices
- Events

How it pays for itself

Small fees, real data.

8%
commission on each booking

Partners
campuses and parking providers

Plans
subscriptions for lot operators

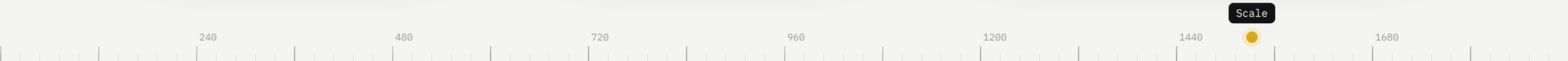
Roadmap

What we build next.

Now
Live plan, booking, pass, lens and 3D on the KIET main lot.

Next
UPI payments, QR check-in at the gate, more KIET lots.

Later
Cameras or sensors for occupancy, and demand forecasts so lots can plan ahead.



15 — Team Arrivo

Who built what.

each of us can demo it.

KS

Krish Singh

Idea & research

explains

The problem, the survey of the lot and the SDG impact.

KS

Kritika Sakhuja

Lot plan & lens

explains

How the plan was traced from the satellite photo, the lens and 3D view.

PS

Prisha Soam

Backend

explains

server.py, the database rule against double booking and live updates.

VK

Vaibhav Kumar

Design & frontend

explains

The site, the booking flow, dark mode and the bay pass.

go on, ask us anything.

Book a bay ↗

Button · 308 × 92

16 —— Thank you

Your turn.

arrivo.

Team Arrivo · INNOTECH'26 · Category 6

Your turn

240 480 720 960 1200 1440 1680

A —— Appendix

Questions we expect.

Q1 What if someone books and never comes?

The bay waits 15 minutes past their arrival time, then opens up for anyone.

Q2 How do you know a bay is really empty?

Today, from bookings. Next, a QR check-in at the gate; later, cameras or sensors.

Q3 Can two people book the same bay?

No. The server checks inside a locked step, and the database has a rule that refuses overlaps.

Q4 What about privacy?

No sign-up. A random device ID, stored only as a hash, and other people's plates are masked.

Q5 Why not just use Google Maps?

Maps tells you a lot exists. Arrivo tells you which bay is free, and holds it for you.

Q6 How does it make money?

A small commission per booking, plus subscription plans for lot operators.

Q7 Do I need to install an app?

No. It's a website that works in any phone browser. The server is one Python file.

Q8 What did you find hardest?

Making the plan match the real lot, and keeping every screen in sync without refreshing.